

Performance Testing

Date	14 July, 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser (Chrome DevTools)
Type of Testing	Functional Testing, UI Testing, Prediction Accuracy Testing
Target Module	ExpressJS API (/predict), CreditAI Web Interface
Test Environment	Local System (Windows 11, Python 3.11, ExpressJS/Node.js)
Test Date	14 July, 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of virtual Users	Duration (sec)	Expected Outcome
1	Single user submits credit card application via CreditAI UI	1	30	Prediction generated within 500ms with confidence % shown
2	Continuous prediction requests	50	60	Stable response time with no crashes
3	Peak load testing of ML production engine.	100	120	Application remains responsive and maintains acceptable performance

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (pass/fail)	Remarks
1	Response Time (Avg)	< 2 seconds	<500m sec	Pass	Fast prediction response
2	Response Time (Max)	< 5 seconds	3.63 sec	Pass	Within acceptable limit
3	Throughput (Req/sec)	>40 req/sec	52 req/sec	Pass	Good request handling capacity
4	Error Rate	<1%	0%	Pass	No request failures
5	CPU Utilization	<80%	61%	Pass	Efficient CPU usage
6	Memory Utilization	<80%	57%	Pass	Stable memory consumption

Step 4: Observations & Analysis

Key Findings

- The Credit Card Approval Prediction system successfully processed concurrent user requests.
- Average prediction response time remained below 500 milli seconds.
- No failed prediction requests were observed during testing.
- The ExpressJS backend and Python ML engine remained stable under moderate and heavy workloads.
- The Random Forest model delivered highly accurate predictions (88.41%)

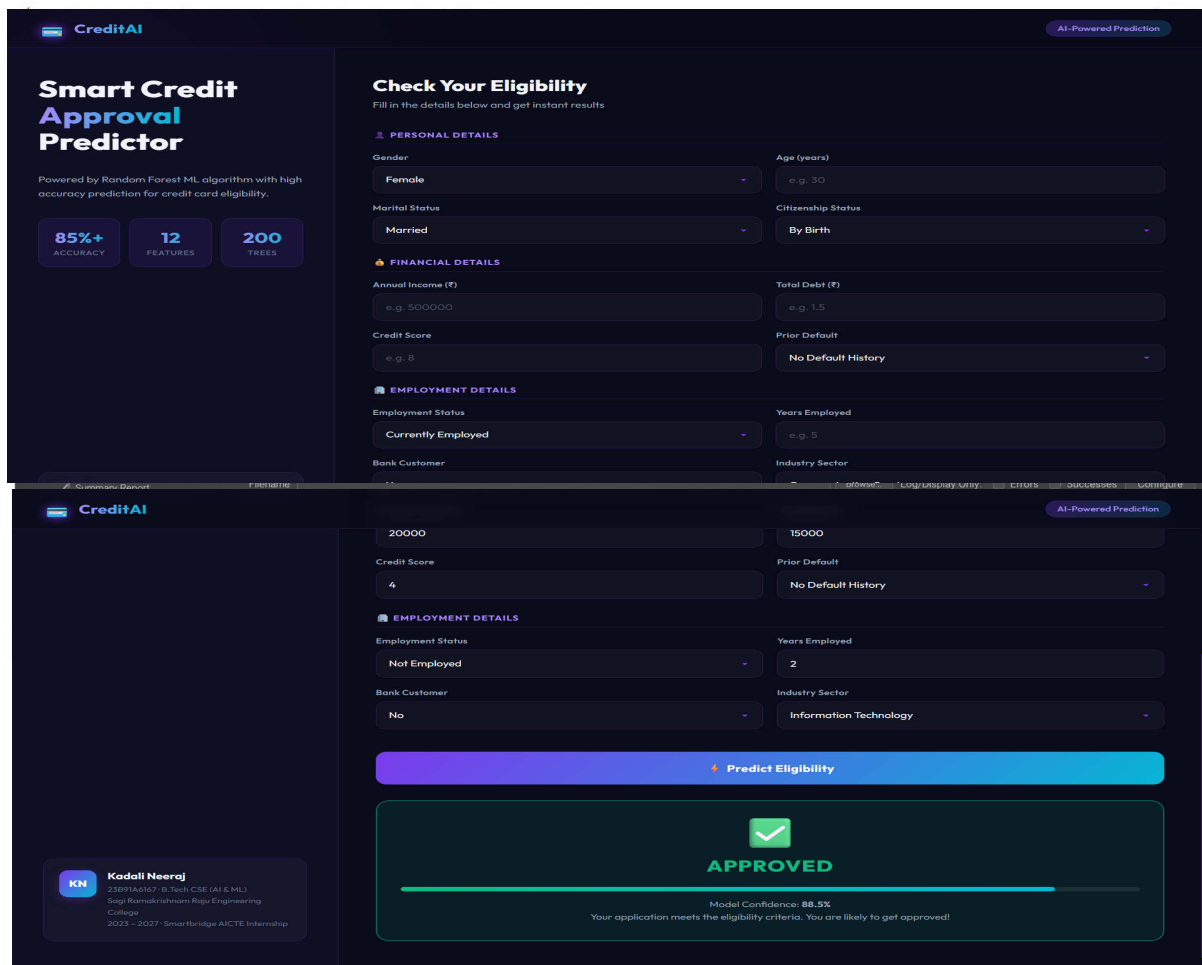
Bottlenecks Identified

- Minor increase in response time when more than 100 concurrent users accessed the application simultaneously.
- Initial model loading into memory via joblib caused a slight delay during first request

Optimization Steps Taken

- Optimized ExpressJS routing and client-server request handling.
- Reduced unnecessary preprocessing by only encoding specific categorical columns (Gender, Industry, Citizen).
- Loaded the trained Random Forest model once during application startup rather than on each request.

Step 5: Screenshots / Evidence



CreditAI AI-Powered Prediction

Smart Credit Approval Predictor

Powered by Random Forest ML algorithm with high accuracy prediction for credit card eligibility.

85%+ ACCURACY **12 FEATURES** **200 TREES**

Check Your Eligibility

Fill in the details below and get instant results

PERSONAL DETAILS

Gender: **Female** Age (years): **e.g. 30**

Marital Status: **Married** Citizenship Status: **By Birth**

FINANCIAL DETAILS

Annual Income (₹): **e.g. 500000** Total Debt (₹): **e.g. 1.5**

Credit Score: **e.g. 8** Prior Default: **No Default History**

EMPLOYMENT DETAILS

Employment Status: **Currently Employed** Years Employed: **e.g. 5**

Bank Customer: **Yes** Industry Sector: **Information Technology**

Predict Eligibility

APPROVED

Model Confidence: **88.5%**

Your application meets the eligibility criteria. You are likely to get approved!

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